Fadi Abidi

Controls and Automation Engineer

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PROFESSIONAL SUMMARY

Controls and Automation Engineer with 6 years of industrial experience across photovoltaic, electronics and automotive manufacturing, specializing in PLC programming, industrial robotics integration, machine vision and full commissioning of automated production systems. Full project lifecycle experience covering electrical design, hardware and software integration, PLC programming, on-site commissioning, troubleshooting and customer training at international client sites. Programming based on IEC 61131-3 standard languages, transferable across PLC platforms. Strong background in lean manufacturing and structured problem solving (8D, PFMEA, Six Sigma). Experienced in reading and interpreting electrical schematics, coordinating with mechanical and electrical engineering teams and supporting customers directly on site.

TECHNICAL SKILLS

PLC Programming: Beckhoff TwinCAT 3, Siemens TIA Portal (S7-1200, S7-1500), Structured Text (ST), Ladder Logic (LD), SFC/Grafcet, Function Block Diagram (FBD), IEC 61131-3

Industrial Robotics: Estun (ER Series), KUKA KR 210, ABB RobotStudio, pick and place, vision-guided robotics, robot-to-PLC coordination

Machine Vision: Hikrobot MVS, Cognex VisionPro, Keyence, camera calibration, inspection routines, robot guidance

HMI and SCADA: TwinCAT HMI, Siemens WinCC, AVEVA InTouch (Wonderware), IX Developer, InoTouchPad, EasyBuilder Pro

Industrial Networks and Bus Systems: EtherCAT, PROFINET, EtherNet/IP, Modbus TCP, TCP/IP, network diagnostics with Wireshark

Drives and Motion Control: Yaskawa servo drives and VFD (SigmaWin+), Inovance VFD, multi-axis motion control, autotuning, fault diagnosis

Electrical and Machine Safety: Electrical schematic reading and interpretation, EPLAN, control cabinet troubleshooting, Beckhoff TwinSAFE, safety interlocks, guarding, CE-marked equipment, Machinery Directive compliance

Commissioning: Electrical and software commissioning, FAT and SAT support, system testing, end-user training, commissioning checklists and documentation

Quality and Problem Solving: PFMEA, Control Plans (VDA/AIAG), 8D, Ishikawa, 5 Whys, MSA, SPC, PPAP/APQP, ECN management, root cause analysis

Lean and Continuous Improvement: VSM, OEE, SMED, 6S, Kaizen, Six Sigma Yellow Belt, WIP reduction, KPI dashboards

Tools and Software: EPLAN, Festo Automation Suite, Wireshark, SolidWorks, Python, Microsoft Excel, Microsoft Office

PROFESSIONAL EXPERIENCE

Automation Engineer, Mondragon Assembly Group

July 2024 - Present | Tunisia and international client sites, on-site

- Commissioned 4 complete photovoltaic production lines and turnkey automated machines at customer sites across the MENA region, Europe and Asia, covering electrical and software commissioning, system testing and end-user training.
- Programmed and configured Beckhoff TwinCAT 3 and Siemens TIA Portal S7 PLCs: SFC/Grafcet sequences, Structured Text, Ladder Logic, Function Block Diagrams, I/O mapping, EtherCAT configuration, machine communication and alarm management.
- Programmed ABB and Estun robots for vision-guided pick and place, integrating robot-to-PLC communication and peripheral equipment.
- Integrated machine vision systems (Hikrobot MVS, Cognex VisionPro, Keyence), from camera calibration to inspection routines and robot guidance.
- Developed HMIs on TwinCAT HMI, WinCC, InTouch, IX Developer, InoTouchPad and EasyBuilder Pro, including screen design, tag mapping and alarm visualization.
- Configured and diagnosed EtherCAT, PROFINET, EtherNet/IP and Modbus TCP networks.
- Tuned and troubleshot Yaskawa and Inovance servo systems and VFDs, from fault diagnosis to autotuning and motion optimization.
- Resolved recurring failures across PLC logic, drives, vision and power supply layers through root cause analysis, turning each fix into a corrective standard for comparable installations.
- Represented the company directly with international customers, working from electrical schematics alongside mechanical, electrical and software teams.

Electronics Process Engineer, Marquardt Tunisia

October 2022 - June 2024 | Manouba, Tunisia, on-site

- Programmed, operated and optimized electronic production equipment: wave and selective soldering, press fitting, conformal coating and CNC milling.
- Reduced downtime on SMT and THT lines through machine troubleshooting and preventive and corrective maintenance.
- Shortened cycle times and cut defects on SMT and THT processes through defect analysis and continuous improvement.
- Resolved recurring production issues with 8D, Ishikawa and 5 Whys root cause analysis.
- Led Engineering Change Notices (ECNs) from feasibility study to prototype validation.
- Implemented new processes and validated production changes with production, quality and R&D teams, tracking process stability through KPIs.

Methods and Process Engineer, Sumitomo Electric Bordnetze SE

January 2021 - October 2022 | Jendouba, Tunisia, on-site

- Reduced downtime on cutting-area machinery by around 30%, targeting setup times, waiting times and outages through structured HAI-C actions.
- Optimized manufacturing flows with Lean tools (VSM, OEE, SMED, 6S, WIP reduction), reducing waste and improving work balance.
- Led PFMEA risk analyses and coordinated preventive and corrective action plans.
- Solved quality and production issues with 8D, Ishikawa and 5 Whys and developed Control Plans to VDA and AIAG standards.
- Led continuous improvement projects and cross-functional initiatives across production.
- Wrote standardized work instructions and trained supervisors and operators.

Final Year Engineering Intern, Automation, Poulina Group Holding

January 2020 - July 2020 | Ben Arous, Tunisia

- Designed and implemented an automated weighing, labeling and packaging system using a Siemens S7-1200 PLC with TIA Portal HMI.
- Mechanical design and conveyor power calculation using SolidWorks.
- Full commissioning including electrical, electronic and mechanical testing.

EDUCATION

Engineering Degree, Electromechanical Engineering: Université Libre de Tunis, September 2017 - September 2020.

Bachelor's Degree, Mechanical Engineering (Industrial Maintenance): ISET Jendouba, September 2013 - July 2017.

CERTIFICATIONS AND TRAINING

Beckhoff TwinCAT 3 Certifications (12 modules): Vision, HMI, EtherCAT, Motion Basics and Intermediate Motion, Object Oriented Programming, TwinSAFE, Source Control, PLC Basics: Beckhoff Automation, September 2024.

EF SET English Certificate, C1 Advanced: English for STEM, U.S. Department of State, August 2024.

Python 101 for Data Science: IBM Cognitive Class, November 2024.

FMEA and Control Plan Moderator: EQM System, May 2022.

Six Sigma Yellow Belt: 6Sigma Study, January 2022.

FMEA and Control Plan (VDA/AIAG): Max-Power Consulting, January 2022.

SPC (Statistical Process Control) and MSA (Measurement System Analysis): Max-Power Consulting, December 2021.

LANGUAGES

Arabic: native **French:** full professional proficiency (C1) **English:** professional working proficiency (C1, EF SET certified)